

Resolution Method in Propositional Logic

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What is resolution?

Definition

- Resolution is an efficient method for searching for a **proof**.
- It is important to become familiar with resolution, because it is widely used in **automatic theorem provers** and it is also the basis of **logic programming**.

Example

Conjunctive Normal Form (CNF)

Definition

A formula is in conjunctive normal form (CNF) iff it is a conjunction of disjunctions of literals.

Table : **CNF** - Examples

Formula	is CNF
$(\neg p \vee q \vee r) \wedge (\neg q \vee r) \wedge (\neg \neg r)$	YES
$(\neg p \vee q \vee r) \wedge ((p \wedge \neg q) \vee r) \wedge (\neg r)$	NOT because $((p \wedge \neg q) \vee r)$ is not a disjunction
$(\neg p \vee q \vee r) \wedge \neg(\neg q \vee r) \wedge (\neg r)$	NOT because $\neg(\neg q \vee r)$ is a negation of a disjunction, instead of a disjunction

Conjunctive Normal Form (CNF)

Theorem

Every formula in propositional logic can be transformed into an equivalent formula in CNF.

To convert an arbitrary formula to a formula in CNF perform the following steps, each of which **preserves logical equivalence**:

Conjunctive Normal Form (CNF)

① Eliminate all operators except for **negation** ($\neg$), **conjunction** ($\wedge$) and **disjunction** ($\vee$) by substituting logically equivalent formulas:

Input		Output	Rule
$p \Rightarrow q$	$\equiv$	$\neg p \vee q$	Material implication
$p \Leftrightarrow q$	$\equiv$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$	Material equivalence
	$\equiv$	$(\neg p \vee q) \wedge (\neg q \vee p)$	Material implication

Conjunctive Normal Form (CNF)

② Push negations inward using De Morgan's laws until they appear only before atomic propositions or atomic propositions preceded by negations.

Input		Output	Rule
$\neg(p \wedge q)$	$\equiv$	$(\neg p \vee \neg q)$	DeMorgan
$\neg(p \vee q)$	$\equiv$	$(\neg p \wedge \neg q)$	DeMorgan

Conjunctive Normal Form (CNF)

③ Eliminate sequences of negations by deleting double negation operators:

Input		Output	Rule
$\neg\neg p$	$\equiv$	p	Double Negation

Conjunctive Normal Form (CNF)

④ The formula now consists of disjunctions and conjunctions of literals. Use the distributive laws to eliminate conjunctions within disjunctions:

Input	Output	Rule
$p \vee (q \wedge r)$	$\equiv (p \vee q) \wedge (p \vee r)$	Distribution

Clashing literals

Two literals are called "clashing" if one literal is the complement of the other. For example: A and $\neg A$, B and $\neg B$, etc.

Notation

A pair of clashing literals are denoted as I and I^c .

That is:

If $I = A$ then $I^c = \neg A$

If $I = \neg A$ then $I^c = A$

Resolution Rule

$$\frac{(C_1 \cup \ell), (C_2 \cup \ell^c)}{(C_1 - \ell) \cup (C_2 - \ell^c)}$$

Where:

- C_1 and C_2 are clauses.
- $C_1 \cup \ell$ is in the knowledge base.
- $C_2 \cup \ell^c$ is in the knowledge base.
- $(C_1 - \ell)$: remove ℓ from clause C_1
- $(C_2 - \ell^c)$: remove ℓ^c from clause C_2
- $(C_1 - \ell) \cup (C_2 - \ell^c)$: Union of two sets $(C_1 - \ell)$ and $(C_2 - \ell^c)$

Resolution Rule

$$\frac{(C_1 \cup \ell), (C_2 \cup \ell^c)}{(C_1 - \ell) \cup (C_2 - \ell^c)}$$

Meaning:

If there $(C_1 \cup \ell)$ and $(C_2 \cup \ell^c)$ in the knowledge base, then we can infer $(C_1 - \ell) \cup (C_2 - \ell^c)$ by adding this new clause into the knowledge base.

Problem

- INPUT:
 - ① Premise: A set of propositional formulas
 - ② Expected Conclusion: a formula
- REQUIREMENT:
 - Prove that the conclusion by be logically derived from the premises.

Proof by Resolution rule

Procedure

- ➊ Add the premises into knowledge base.
- ➋ Add the negation of the conclusion into the knowledge base.
- ➌ Convert formulas in the knowledge base into clause form.
- ➍ Apply resolution rule constantly to the knowledge base to derive an empty clause, using resolution algorithm.

Resolution Algorithm

Input: S is set of clauses.

Output: S is satisfiable or unsatisfiable.

- ① Define $i = 0$ $S_i = S$.
- ② Choose a pair of clashing clauses C_1, C_2 in S_i that has not been chosen before.
- ③ Apply the resolution rule to C_1, C_2 to obtain new clause C .
- ④ Check for terminated condition if one the two given below is satisfied:
 - S is unsatisfiable: If C is an empty clause then exit
 - S is satisfiable: If all pairs of clashing clauses have be resolved.
- ⑤ If not terminated
 - If C is not a trivial clause, let $i=i+1$, and compute $S_{i+1} = S_i \cup C$; Otherwise, $S_{i+1} = S_i$.
 - Goto Step ②

Example

- Premises:
 - ① A
 - ② $A \Rightarrow B$
- Expected conclusion:
 - ① B

Proof by Resolution rule

Knowledge base contains:

- ① A
- ② $A \Rightarrow B$
- ③ $\neg B$

Convert to clause form:

A	$\equiv$	A
$A \Rightarrow B$	$\equiv$	$\neg A \vee B$
B	$\equiv$	$\neg B$

Proof by Resolution rule

Resolution procedure:

No.	Clauses	Notes
1.	A	Premise
2.	$\neg A \vee B$	Premise
3.	$\neg B$	Premise
4.	B	1, 2
4.	$\square$	3, 4

Example

- Premises:

- ① $Q \Rightarrow (R \Rightarrow S)$
- ② $Q \Rightarrow (T \Rightarrow \neg U)$
- ③ $U \Rightarrow (R \vee T)$

- Expected conclusion:

- ① $Q \Rightarrow (U \Rightarrow S)$

Proof by Resolution rule

Knowledge base contains:

- ① $Q \Rightarrow (R \Rightarrow S)$
- ② $Q \Rightarrow (T \Rightarrow \neg U)$
- ③ $U \Rightarrow (R \vee T)$
- ④ $\neg(Q \Rightarrow (U \Rightarrow S))$

Convert to clause form:

$$\begin{aligned} Q \Rightarrow (R \Rightarrow S) &\equiv \neg Q \vee (\neg R \vee S) \\ &\equiv \neg Q \vee \neg R \vee S \\ Q \Rightarrow (T \Rightarrow \neg U) &\equiv \neg Q \vee (\neg T \vee \neg U) \\ &\equiv \neg Q \vee \neg T \vee \neg U \\ U \Rightarrow (R \vee T) &\equiv \neg U \vee (R \vee T) \\ &\equiv \neg U \vee R \vee T \\ \neg(Q \Rightarrow (U \Rightarrow S)) &\equiv \neg(\neg Q \vee (\neg U \vee S)) \\ &\equiv \neg(\neg Q \vee \neg U \vee S) \\ &\equiv \neg\neg Q \wedge \neg\neg U \wedge \neg S \\ &\equiv Q \wedge U \wedge \neg S \end{aligned}$$

Proof by Resolution rule

Resolution procedure:

No.	Clauses	Notes
1.	$\neg Q \vee \neg R \vee S$	Premise
2.	$\neg Q \vee \neg T \vee \neg U$	Premise
3.	$\neg U \vee R \vee T$	Premise
4.	Q	Premise
5.	U	Premise
6.	$\neg S$	Premise
7.	$\neg R \vee S$	1, 4
8.	$\neg R$	6, 7
9.	$R \vee T$	3, 5
10.	$\neg T \vee \neg U$	2, 4
11.	$\neg T$	5, 10
12.	R	9, 11
	$\square$	8, 12, Proved

Example

- Premises:

- ① $(N \vee O) \Rightarrow (C \wedge D)$

- ② $(D \vee K) \Rightarrow (P \vee \neg C)$

- ③ $(P \vee G) \Rightarrow \neg(N \wedge D)$

- Expected conclusion:

- ① $\neg N$

Proof by Resolution rule

Knowledge base contains:

- ① $(N \vee O) \Rightarrow (C \wedge D)$
- ② $(D \vee K) \Rightarrow (P \vee \neg C)$
- ③ $(P \vee G) \Rightarrow \neg(N \wedge D)$
- ④ N

Proof by Resolution rule

Convert to clause form:

$$\begin{aligned}(N \vee O) \Rightarrow (C \wedge D) &\equiv \neg(N \vee O) \vee (C \wedge D) \\ &\equiv (\neg N \wedge \neg O) \vee (C \wedge D) \\ &\equiv (\neg N \vee C) \wedge (\neg N \vee D) \wedge \\ &\quad (\neg O \vee C) \wedge (\neg O \vee D) \\ (D \vee K) \Rightarrow (P \vee \neg C) &\equiv \neg(D \vee K) \vee (P \vee \neg C) \\ &\equiv (\neg D \wedge \neg K) \vee (P \vee \neg C) \\ &\equiv (\neg D \vee P \vee \neg C) \wedge (\neg K \vee P \vee \neg C) \\ (P \vee G) \Rightarrow \neg(N \wedge D) &\equiv \neg(P \vee G) \vee \neg(N \wedge D) \\ &\equiv (\neg P \wedge \neg G) \vee (\neg N \vee \neg D) \\ &\equiv (\neg P \vee \neg N \vee \neg D) \wedge (\neg G \vee \neg N \vee \neg D) \\ N &\equiv N\end{aligned}$$

Proof by Resolution rule

Resolution procedure:

No.	Clauses	Notes
1.	$\neg N \vee C$	Premise
2.	$\neg N \vee D$	Premise
3.	$\neg O \vee C$	Premise
4.	$\neg O \vee D$	Premise
5.	$\neg D \vee P \vee \neg C$	Premise
6.	$\neg K \vee P \vee \neg C$	Premise
7.	$\neg P \vee \neg N \vee \neg D$	Premise
8.	$\neg G \vee \neg N \vee \neg D$	Premise
9.	N	Premise

No.	Clauses	Notes
10.	C	1, 9
11.	$\neg D \vee P$	5, 10
12.	D	2, 9
13.	P	11, 12
14.	$\neg P \vee \neg D$	7, 9
15.	$\neg P$	12, 14
	$\square$	13, 15, Proved